

For reference when drafting articles, please refer to the final report for accurate analysis methods and parameters.

Methods for Visium HD Product

Visium HD Experiment and Sequencing

Sample preparation (For fresh frozen sample)

Fresh tissues were concurrently frozen and embedded in optical cutting tissue (OCT) compound in liquid nitrogen. The RNA quality of OCT embedded block was assessed by Agilent 2100. RNA integrity number (RIN) of tissues greater than 4 were used for visium spatial gene expression experiments. Cryosections were performed on a Leica CM3050 and brightfield images were taken on a Leica Aperio Versa8 whole-slide scanner at 20× resolution.

Sample preparation (for FFPE sample)

According to the Visium HD FFPE Tissue Preparation Handbook (CG000684, 10x Genomics), small pieces of tissue were formalin fixed and paraffin embedded (FFPE), and RNA quality assessment of FFPE tissue blocks was performed by calculating the percentage of total RNA fragments >200 nucleotides (DV200) of RNA extracted from tissue sections. For the high RNA quality, DV200 > 30% was required.

Visium Sequencing Libraries Preparation (For fresh frozen sample)

Tissue block preparation, sectioning, H&E staining, and imaging followed the Visium HD Fresh Frozen Tissue Preparation Handbook (CG000763). Probe hybridization, probe ligation, slide preparation, probe release, extension, and library construction followed the Visium HD Spatial Gene Expression Reagent Kits User Guide (CG000685).

Visium Sequencing Libraries Preparation (For FFPE sample)

FFPE tissue sections were placed on blank slides for deparaffinization, HE staining, and imaging. Based on the Visium HD Spatial Gene Expression Reagent Kits User Guide (CG000685), probe hybridization, probe ligation, Visium HD slide preparation, probe release, extension and probe-based library construction were carried out.

Sequencing

Sequencing was performed with a Novaseq Xplus PE150 platform according to the manufacturer's instructions (Illumina) with paired-end reads.

Primary Bioinformatical Analysis

Data processing

We utilize an in-house script to conduct basic statistical analyses on raw data, assessing both data quality and GC content throughout the sequencing cycles. The raw FASTQ files and histology images were processed on a per-sample basis using the Space Ranger software (version 3.0.0, 10X Genomics) with default settings. Subsequently, the filtered gene-bins matrix and the fiducial-aligned, low-resolution images were employed for downstream data analyses using Scanpy.

Scanpy analysis

The Python package Scanpy (version 1.10.1) was utilized to carry out gene expression normalization, dimensionality reduction, spot clustering, and differential expression analysis. Briefly, the analysis was conducted using the expression matrix derived from 16 μ m bins. Bins were filtered to include only those with a minimum detected gene count of 100. Normalization across spots was executed using the `scanpy.pp.normalize_total` function, and 3,000 highly variable genes were selected for subsequent principal component analysis. For spot clustering, the first five principal components were utilized to construct a graph, which was then segmented with a resolution of 0.1. Differential gene expression analysis for each cluster was performed using the Wilcoxon rank-sum test via the `scanpy.tl.rank_genes_groups` function. Genes were considered significantly differentially expressed if they exhibited a fold change greater than 0 and an adjusted p-value less than 0.05.

Enrichment analysis

The clusterProfiler R package (version 3.10.1) was utilized to conduct enrichment tests for candidate gene sets based on the hypergeometric distribution. Pathways with a corrected p-value less than 0.05 were deemed significantly enriched. Three pathway classification systems were employed as reference databases for human or mouse genes: Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and Reactome Pathway. GO and Reactome annotations were retrieved from Ensembl BioMart, while KEGG pathway annotations were obtained via the KEGG REST API.